Technical Knowledge Base Article

Reference Pipeline for EXIF Integrity Preservation and 24-Hour Face Retention

AI Platform Implementation Engineer Assessment

Prepared by: Rishi Oberoi

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Executive summary. This solution addresses the two explicit controls in the brief: preserving EXIF metadata required for downstream model functionality, and enforcing 24-hour hard deletion for images containing human faces. The pipeline validates metadata at source and after transfer, reproduces the bridge stripping failure mode in controlled test cases, restores required EXIF fields when stripping occurs, and creates retention records for detected face images. Additional testing with dark and blurry warehouse worker images confirmed the quality issue described in the escalation scenario and informed the remediation plan presented later in this article.
GitHub Repository Link: https://github.com/rishioberoi-dev/ai-platform-assessment

Objective	**Fix EXIF metadata loss, enforce 24-hour face retention, and document investigative findings for the warehouse image quality issue.**
Assessment framing	Part 1 is solved in code; Part 2 is addressed through representative dark/blurry worker image testing and a remediation plan.
Evidence sources	Final terminal run, manifests, retention records, audit logs, and three controlled bridge-strip demonstration images.
Final run summary	Processed: 19 | Approved: 14 | Restricted face retention: 5 | EXIF preserved: 19 | EXIF lost in processing after repair: 0

1. Problem Summary

1.1 Part 1: Mandatory controls from the brief

The brief requires a solution that fixes two concrete operational risks. First, the Transfer Bridge strips EXIF metadata, specifically GPS and timestamp context needed for the model to function reliably. Second, every image containing a human face must be hard deleted within 24 hours. The pipeline therefore had to preserve metadata integrity through transfer and isolate detected face images into a restricted retention workflow.

1.2 Part 2: Escalation scenario

The escalation describes poor model quality after production images shifted from bright office conditions to darker and blurrier warehouse conditions. To investigate that issue, representative warehouse worker images

were included in the final test batch. Those images were intentionally renamed to reflect the operational scenario and used as evidence that the team has started validating the failure mode described by leadership.

2. Solution Architecture

The reference implementation follows a simple but defensible design: validate metadata at source, classify privacy-sensitive images, transfer the file, verify metadata again, and restore required EXIF fields if the transfer step strips them. The original source file is treated as the metadata source of truth, while manifests and audit logs preserve a traceable record of every decision.

Control point	Why it exists	Implemented logic	Evidence produced
Source EXIF validation	Confirm whether required GPS/timestamp data exists before transfer.	Read source EXIF, summarise timestamp and GPS state, preserve required fields as a system record.	Source EXIF scan result in terminal + source_exif_summary in manifest.
Face retention routing	Meet the 24-hour hard deletion requirement for face images.	Run DNN-based face detection. If face detected, route to restricted_face_retentio n and generate retention record.	Terminal routing output + *_retention.json manifest.
Post-transfer EXIF validation	Detect whether the bridge stripped metadata during transfer.	Re-read EXIF after copy/transfer and compare source vs processed state.	Processed EXIF scan result + integrity comparison in terminal.
Metadata restoration	Repair bridge stripping when source EXIF existed but processed EXIF was lost.	Reapply preserved EXIF fields to the processed artifact and revalidate.	Bridge-strip demo output + restored EXIF evidence in terminal and manifest.

3. Operational Flow

The terminal output was intentionally designed to mirror the system flow so that an operator or reviewer can follow the order of operations in a live walkthrough.

Step	Stage	Operational action
1	Scan source image	Read the source image and validate EXIF metadata, printing timestamp and GPS values to the terminal.
2	Check privacy rule	Run face detection and determine whether the image must enter the restricted 24-hour retention path.
3	Route image	Move the image to approved or restricted_face_retention based on face detection outcome.
4	Validate moved image	Re-read EXIF metadata after transfer to confirm whether metadata survived translation.

5	Compare integrity	Compare source and processed metadata values. If the bridge-strip demo is triggered, the comparison fails by design.
6	Restore if required	If EXIF was lost during transfer, restore required EXIF fields from the preserved source record and revalidate the file.
7	Record evidence	Write manifest, update audit log, and, for face images, generate a retention record for 24-hour deletion handling.

4. Key Code Logic

The code below captures the two most important behaviours in the solution: integrity verification across transfer, and metadata restoration when the bridge strips EXIF.

4.1 EXIF integrity verification

```
def determine_exif_integrity(source_exif_summary, processed_exif_summary):
    source_has_required = source_exif_summary["required_context_present"]
    processed_has_required = processed_exif_summary["required_context_present"]

    if source_has_required and processed_has_required:
        return ("exif_preserved", "EXIF metadata integrity verification completed successfully.")

    if not source_has_required and not processed_has_required:
        return (
            "exif_preserved",
            "EXIF metadata integrity verification completed successfully. Source and processed image
match, although required GPS/timestamp context was absent at source.",
        )

    return (
        "exif_lost_in_processing",
        "EXIF metadata integrity verification failed: required metadata was present in the source image
but missing after processing.",
    )
```

4.2 Bridge-strip simulation and repair path

```
bridge_strip_simulated = simulate_bridge_exif_stripping(processed_path)

processed_exif_data = extract_exif_metadata(processed_path)
processed_exif_summary = summarize_exif_status(processed_exif_data)
exif_integrity_status, exif_integrity_message = determine_exif_integrity(
    source_exif_summary,
    processed_exif_summary,
)

if exif_integrity_status == "exif_lost_in_processing":
    metadata_restoration_applied = True
    restore_ok = restore_required_exif_fields(processed_path, preserved_required_exif)

    processed_exif_data = extract_exif_metadata(processed_path)
    processed_exif_summary = summarize_exif_status(processed_exif_data)
    exif_integrity_status, exif_integrity_message = determine_exif_integrity(
        source_exif_summary,
        processed_exif_summary,
```

```
)
```

5. Test Evidence from Final Run

The final run processed 19 images. Three bridge-strip demonstration files reproduced the transfer bug and confirmed the repair path. Five images entered the restricted face retention path. Three dark/blurry warehouse worker images were retained as investigative evidence for the Part 2 quality problem and were not detected as face-containing by the current detector, which is consistent with leadership's concern that warehouse imagery is materially harder for the model than bright office imagery.

Image	Scenario	Status	EXIF result	What it proves
bridge_strip_demo_1.jpg	Controlled bridge bug	approved	exif_preserved after repair	Bridge stripping reproduced and repaired successfully.
bridge_strip_demo_2.jpg	Controlled bridge bug	approved	exif_preserved after repair	Repair path is repeatable, not a one-off case.
bridge_strip_demo_3.jpg	Controlled bridge bug	approved	exif_preserved after repair	Third confirmation of the bridge restoration path.
office-12.jpg	Office image with face	restricted_face_retention	exif_preserved	Privacy rule routes detected face images into 24-hour retention flow.
office-3.jpg	Office image with face	restricted_face_retention	exif_preserved	Retention record creation works in normal face-detection cases.
blurry-warehouse.jpg	Blurry warehouse worker image	approved	exif_preserved	Investigative evidence for Part 2: warehouse imagery can evade current face detection.
dark-warehouse.jpg	Dark warehouse worker image	approved	exif_preserved relative to source	Investigative evidence for Part 2 and basis for the remediation plan.
dark-blurry-warehouse-with-worker.png	Dark/blurry worker image	approved	exif_preserved relative to source	Confirms the same failure mode under weaker image conditions.

6. Value Delivered by the Solution

- Introduced an explicit before-and-after EXIF integrity control so metadata failures are visible rather than silent.
- Added a recovery path that restores required EXIF metadata onto the processed artifact when the transfer step strips it.
- Implemented a restricted retention workflow with a generated 24-hour deletion record for detected face images.
- Produced traceable operational evidence through manifests and audit logs for every processed file.
- Started testing against dark/blurry warehouse worker imagery, creating concrete evidence for the Part 2 remediation plan rather than leaving the escalation theoretical.

7. Findings Relevant to the CEO / VP Escalation

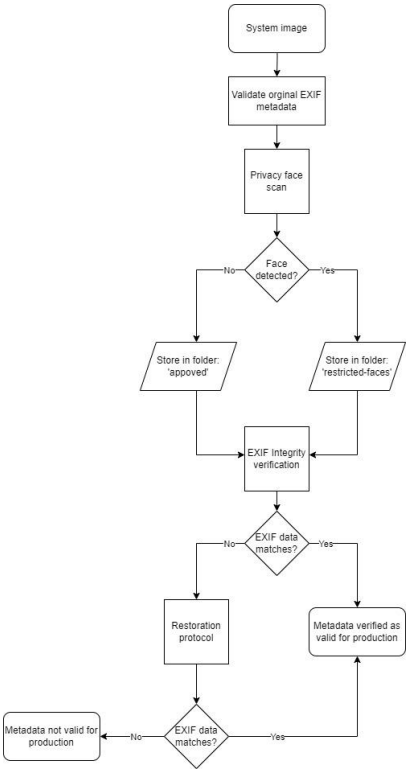
The three dark/blurry warehouse worker images were intentionally retained in the final test batch because they represent the production conditions described in the escalation scenario. All three were approved rather than entering the restricted face path, which demonstrates that the team has reproduced the quality issue described by leadership: the current detector performs materially better on brighter office-style imagery than on dark/blurry warehouse-style imagery. This means the Part 1 controls are in place, but Part 2 requires a targeted improvement plan for warehouse imagery.

7.1 Proposed remediation plan for Part 2

Action	Purpose	Expected value
Curate warehouse test set	Build a representative dataset of dark, blurry, oblique-angle worker images.	Moves model tuning from anecdotal failure to measurable test coverage.
Tune detector threshold and review path	Adjust confidence thresholds and introduce manual review for uncertain cases.	Reduces the risk of false negatives on worker faces in poor conditions.
Retrain or augment on warehouse conditions	Expand training coverage beyond bright office conditions.	Improves robustness where the current model underperforms according to the escalation.
Integrate the same EXIF controls into the real bridge	Carry the proof-of-concept logic into the production transfer path.	Ensures the metadata-loss fix is not isolated to the local reference pipeline.

8. Draw.io Flowchart

The flowchart accompanying this document mirrors the terminal logic so that the system chart and runtime evidence tell the same story. The flow is: Start → Scan source image → Validate source EXIF metadata → Detect human face? → Route to approved or restricted_face_retention → Move image → Validate EXIF on moved image → Compare source vs processed metadata → If EXIF lost, restore metadata from preserved source record → Revalidate EXIF → Create retention record if face detected → Write manifest → Update audit log → Final result.



9. Conclusion

The reference pipeline solves the core Part 1 problem by preserving metadata integrity across transfer, repairing the bridge stripping failure mode in controlled test cases, and enforcing a dedicated 24-hour retention workflow for detected face images. The final warehouse worker tests also provide useful Part 2 evidence: the team has already begun validating the harder warehouse conditions that are driving the reported model-quality issues, and the remediation plan is therefore based on observed behaviour rather than assumption.